



**SILVER OAK
UNIVERSITY**
EDUCATION TO INNOVATION



SILVER OAK UNIVERSITY

**School of Technology, Design and Computer
Application**

**Bachelor of Computer Application / Bachelor of Science
Computer Science & Information Technology / Integrated
Master of Science Information Technology / Bachelor of
Science Data Science**

Course Name: Software Testing

Course Code: SEC2013C

Semester: 3rd

Prerequisite:

1. Basic understanding of computer systems, software applications, and web technologies.
2. Fundamental knowledge of programming concepts and software development processes.
3. Familiarity with web browsers, websites, and client-server applications.
4. Basic analytical and logical thinking skills for identifying software defects and designing test cases.
5. Awareness of software quality concepts and debugging practices.

Course Objectives:

1. Understand software testing concepts, SDLC/STLC phases, and testing documentation.
2. Apply manual testing techniques, test case design methods, and defect reporting practices.
3. Analyze software applications using functional, non-functional, integration, system, and regression testing techniques.
4. Develop automation testing solutions using Python and Selenium WebDriver.
5. Utilize industry-standard testing tools for defect management, test execution, and reporting.
6. Create professional testing artifacts such as Test Plans, RTM, Test Reports, and Automation Test Scripts.

Teaching Scheme:

Teaching & Learning Scheme (In Hrs Per Semester)									Total Credits
Classroom Instruction						Extended Learning		Total Hours	
L		T		P					
Cr	Hr	Cr	Hr	Cr	Hr	TW	SL	Hr	C
0	0	2	30	0	0	15	15	60	2

List of Tutorials:

Sr. No.	Tutorial Content	Hours	Co No.
1	Understand SDLC and STLC phases for a real-world software project and identify testing activities performed during each development stage.	2	CO-1
2	Understand a professional test plan, Requirement Traceability Matrix (RTM), testing scope document, and testing strategy for a sample application.	2	CO-1
3	Understand requirement reviews, code walkthroughs, inspections, and peer review activities using sample project documents.	2	CO-1
4	Apply functional, non-functional, positive, and negative test cases for modules such as login, registration, and payment systems.	2	CO-2

5	Apply defect identification and prepare professional bug reports including severity, priority, screenshots, and reproduction steps.	2	CO-2
6	Apply manual test cases using multiple input conditions and prepare detailed test execution and summary reports.	2	CO-2
7	Apply Boundary Value Analysis (BVA) and Equivalence Partitioning techniques for validation testing of forms and authentication modules.	2	CO-2
8	Apply defect management, test case management, and execution report generation using Bugzilla and TestLink workflows.	2	CO-2
9	Analyze black-box testing, white-box testing, integration testing, and system testing on a sample web application.	2	CO-3
10	Analyze project modules and identify suitable functionalities for automation testing and regression testing implementation.	2	CO-3
11	Analyze automation testing in Python for website navigation, forms, multi-page workflows, links, and regression testing scenarios.	2	CO-3
12	Analyze AI-powered software testing tools and apply Generative AI techniques for test case generation, test data creation, defect prediction, requirement analysis, and automated Selenium test script generation for software quality assurance.	2	CO-3
13	Create and configure Python, Selenium WebDriver, VS Code/PyCharm IDE, browser drivers, and automation testing libraries for Selenium testing.	2	CO-4
14	Create Selenium automation scripts in Python for browser navigation, form handling, web element interaction, and validation testing.	2	CO-4
15	Create automated login functionality, authentication validation, assertions, and exception handling scenarios using Selenium WebDriver with Python.	2	CO-4
Total Hours		30	

Course Outcomes:

CO No.	CO Statement	Blooms Taxonomy Level	Tutorials
CO-1	Explain software testing concepts, SDLC/STLC phases, and testing documentation.	Understand	1, 2, 3
CO-2	Apply manual testing, test case design, defect reporting, and validation techniques.	Apply	4, 5, 6, 7, 8
CO-3	Analyze software modules using functional, non-functional, integration, system, automation testing, and AI-assisted testing techniques.	Analyze	9, 10, 11, 12
CO-4	Develop Selenium automation scripts using Python for web application testing and regression testing.	Create	13, 14, 15

Teaching & Learning Pedagogy approaches:

- Problem-Based Learning: Real-world software testing problems for analytical and testing skills development.
- Design Thinking: Structured approach for designing test cases, scenarios, and defect reports.
- Cooperative-Based Learning: Group activities for peer reviews, debugging, and testing exercises.
- Competency-Based Learning: Skill development in manual and automation testing using testing tools.
- Experiential Learning: Hands-on practice in test execution, reporting, and Selenium automation.
- Industry-Oriented Learning: Case studies and mini projects using real-world testing practices.
- ICT-Based Learning: Use of modern tools and IDEs for technology-enabled practical learning.

Major Equipment and Simulation Tools:

1. Computer Systems with Internet Facility
2. Python Programming Environment
3. Selenium WebDriver
4. VS Code / PyCharm IDE
5. Web Browsers (Chrome, Firefox, Edge)
6. Browser Drivers (ChromeDriver, GeckoDriver)
7. Bugzilla / TestLink Tools
8. Sample Web Applications for Testing
9. Projector and Smart Classroom Facilities
10. Software Testing and Automation Libraries

Books Recommended:

1. Graham, D., Black, R., & van Veenendaal, Foundations of software testing: ISTQB certification. Cengage Learning.
2. Thompson, G., Morgan, P., & Samaroo, Software testing: An ISTQB-BCS certified tester foundation level guide. BCS Learning & Development Ltd.
3. Jigjinni, V. K. Learning Selenium testing tools with Python. Packt Publishing.
4. Avasarala, S. Selenium WebDriver practical guide. Packt Publishing.
5. Rasool, Y. Automated software testing with Python. Packt Publishing.
6. Das, A., & Bhattacharya, D. Hands-on Selenium WebDriver with Python. BPB Publications.

Term Work Module:

15 HOURS

Objective: The Term Work Module aims to provide practical understanding of software testing concepts, testing documentation, and manual testing techniques through hands-on activities and real-world testing scenarios. It also helps students analyze different testing approaches and develop Selenium automation scripts using Python for web application testing, validation, and regression testing.

Term Work	Statement	Hours	CO No.
Term Work 1	Understand SDLC and STLC phases, testing activities, requirement reviews, and prepare basic testing documents such as Test Plan, RTM, and Testing Strategy for a sample software project.	3 Hours	CO-1

Term Work 2	Apply functional, non-functional, positive, and negative test cases for modules like login and registration, and prepare professional bug reports with severity, priority, screenshots, and reproduction steps.	2 Hours	CO-2
Term Work 3	Apply Boundary Value Analysis (BVA), Equivalence Partitioning, manual test execution, and defect management activities using Bugzilla and TestLink for sample applications.	3 Hours	CO-2
Term Work 4	Analyze black-box testing, white-box testing, integration testing, system testing, and explore AI-powered software testing tools for intelligent test case generation, defect prediction, requirement analysis, and automated test script generation for software quality assurance.	2 Hours	CO-3
Term Work 5	Create and configure Selenium testing environment using Python, VS Code/PyCharm, browser drivers, and develop Selenium automation scripts for browser navigation, login authentication, form handling, assertions, validation, and exception handling.	5 Hours	CO-4
Total Hours		15	

Self Learning Module:

15 HOURS

Objective: The Self Learning Module aims to help students understand software testing concepts, testing documentation, manual testing techniques, defect management, and modern AI-assisted testing practices through guided self-study activities and online resources. It also enables students to develop practical skills in Selenium automation testing using Python, intelligent test case generation, automated test script creation, and software quality assurance.

Recommended Online Resources:

Sr. No.	Topic Name	Resource Link
1	Study SDLC and STLC phases and prepare a basic Test Plan, RTM, and Testing Strategy document for a sample web application.	NPTEL Software Testing Course Guru99 Software Testing Tutorial
2	Design functional, non-functional, positive, and negative test cases for Login, Registration, and Payment modules.	ISTQB Official Website TutorialsPoint Manual Testing Tutorial
3	Practice defect reporting with severity and priority classification and explore Bugzilla and TestLink workflows.	Bugzilla Official Website TestLink Official Website

4	Study Black-box Testing, White-box Testing, Integration Testing, System Testing, Regression Testing, and explore AI-powered testing tools for intelligent test case generation, automated test script creation, visual testing, and software quality assurance.	Selenium Official Documentation: https://www.selenium.dev/documentation/?utm_source=.com OpenAI Platform Documentation: https://developers.openai.com/api/docs Applitools Documentation: https://applitools.com/docs/index.html?utm_source=.com Testim Documentation: https://docs.tricentis.com/testim/content/overview/testim-overview/index.htm ACCELQ Documentation: https://support.accelq.com/hc/en-us/articles/115010298928-What-is-ACCELQ?utm_source=.com
5	Install Python, Selenium WebDriver, VS Code/PyCharm, configure browser drivers, and develop basic Selenium scripts for login automation and form validation.	Python Official Website Selenium with Python Documentation Visual Studio Code

CO-PO-PSO Matrix:

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2
CO-1	3	2	-	1	-	-	2	-
CO-2	2	3	2	2	1	-	3	1
CO-3	2	3	2	3	2	-	3	2
CO-4	1	2	3	2	3	1	2	3

CO-PO-PSO MAPPING JUSTIFICATION FOR Software Testing

CO No.	CO Statement	PO/PSO	Justification
CO-1	Explain software testing concepts, SDLC/STLC phases, and testing documentation.	PO1	Students acquire fundamental knowledge of software testing processes and quality assurance concepts.
		PO2	Students understand requirement analysis and testing activities throughout SDLC and STLC.
		PO4	Students learn testing investigations, reviews, and documentation practices.

		PSO1	Supports understanding of software quality assurance methodologies and testing principles.
CO-2	Apply manual testing, test case design, defect reporting, and validation techniques.	PO1	Applies software testing fundamentals for practical testing activities.
		PO2	Analyzes requirements and designs suitable test cases for validation.
		PO3	Develops structured test scenarios and testing artifacts.
		PO4	Investigates software defects and performs validation testing.
		PO5	Uses Bugzilla, TestLink, and related testing tools effectively.
		PSO1	Strengthens software quality assurance and defect management skills.
		PSO2	Introduces industry-standard testing tools and platforms.
CO-3	Analyze software modules using functional, non-functional, integration, system, automation testing, and AI-assisted testing techniques.	PO1	Applies software engineering and testing concepts.
		PO2	Analyzes software modules and testing requirements.
		PO3	Evaluates suitable testing strategies and automation approaches.
		PO4	Conducts systematic testing investigations and quality assessment.
		PO5	Utilizes modern testing and AI-powered tools.
		PSO1	Enhances software quality assurance and validation capabilities.

		PSO2	Integrates automation and emerging technologies into testing workflows.
CO-4	Develop Selenium automation scripts using Python for web application testing and regression testing.	PO1	Applies software testing and programming fundamentals.
		PO2	Identifies automation requirements and testing scenarios.
		PO3	Designs and develops Selenium automation solutions.
		PO4	Evaluates automated test execution results and reports.
		PO5	Uses Selenium WebDriver, Python, IDEs, and browser automation tools.
		PO6	Demonstrates professional reporting, documentation, and teamwork practices.
		PSO1	Supports software quality assurance through automation testing.
		PSO2	Applies modern automation technologies for real-world software testing.

MAPPING WITH SUSTAINABLE DEVELOPMENT GOALS

SDG Mapping for Course Name(SEC2013C)		
SDG No and Title	Relevance to Course	Justification
SDG 4 – Quality Education	High	The course develops industry-relevant software testing and automation skills through practical learning, Selenium automation, defect management, test case design, and real-world testing projects. It enhances students' employability and technical competency in software quality assurance.
SDG 8 – Decent Work and Economic Growth	Medium	Software testing is a critical IT industry skill. The course prepares students for careers as Software Test Engineers, QA Engineers, Automation Test Engineers, and Quality Analysts by providing hands-on experience with industry-standard tools such as Selenium, Python, Bugzilla, and TestLink.
SDG 9 – Industry, Innovation and Infrastructure	High	The course contributes to the development of reliable, secure, and high-quality software systems through testing, validation, automation, and defect management practices. These skills support innovation and the creation of robust digital infrastructure.

Explanation:

The Software Testing course introduces students to the principles, techniques, and tools used to ensure software quality and reliability. It covers SDLC and STLC phases, testing documentation, and quality assurance practices. Students learn to design and execute manual test cases, identify defects, and prepare professional bug reports. The course explores various testing approaches including functional, non-functional, integration, system, and regression testing. Students gain hands-on experience with defect management tools and test management processes. The course also focuses on automation testing using Python and Selenium WebDriver. Through practical exercises and real-world testing scenarios, students develop industry-relevant software testing skills. The course prepares learners for careers in software quality assurance, manual testing, and automation testing.